

Guanlin WU

Email: 120010048@link.cuhk.edu.cn | Tel: (+86) 18753542673

Personal Website Link: <https://guanlinwu123.github.io>

EDUCATION

Chinese University of Hong Kong, Shenzhen

2020.09 - Current

- Computer Science and Engineering
- MGPA: 3.470/4.0
- Core Courses: *Data Structure, Computer Architecture, Operating System, Software Engineering, Parallel Computing, Design and Analysis of Algorithm, Optimization, Stochastic Process, Reinforcement Learning*

UC Berkeley

2018.06-2018.08

Academic Talent Development Program

Berkeley, CA

RESEARCH PROJECTS

University of California, Los Angeles/UAV Path Planning and Dynamic Obstacle Avoidance

2023.06 – 2023.09

Research Assistant

Online

- Designed a new UAV system based on 2D-LiDAR that included the Map Construction (based on Hector SLAM), the Path Planner (using RRT), and the Dynamic Obstacle Handler (using D3QN).
- Developed algorithms for LiDAR detection (with noise decreasing and motion trajectory) and Reset in RL training.
- Implemented a new training strategy for the RL model, where Dynamic Obstacle Handler was directly trained in the ROS-Gazebo-PX4 simulator with the UAV model "iris_rplidar."
- Publication: [Guanlin Wu, Zhuokai Zhao, Yutao He. "Reinforcement Learning Enabled 2D-LiDAR Autonomous System for Parsimonious UAVs." Submitted.](#)

SENIOR/AGV Resource Pool

2023.02 – 2023.06

JAVA Engineer

Beijing, China

- Implemented several technical approaches to implement the AGV Resource Pool, including the KM Matching algorithm, Dijkstra with time window, and Dynamic Merging.
- Developed a simulator based on OpenTCS for performance testing with various possible algorithms.

East China Normal University/Virtual Machine Allocation

2022.05 – 2022.11

Research Assistant

Shanghai, China

- Improved Best Fit Algorithm slightly for specific type of DVBP problem by using an innovative way of constructing the Balanced Binary Tree.
- Managed to use Approximation algorithm to prove the ratio between modified Best Fit Algorithm and Optimal solution:

$$\text{BFT}(L) < \frac{20}{9} \text{OPT}(L)$$

- Found the Virtual Machine data about their type from VMagent and HUAWEI.
- Publication: [Lifeng Guo, Changhong Lu, and Guanlin Wu. "Approximation algorithms for a virtual machine allocation problem with finite types." *Information Processing Letters* 180 \(2023\): 106339 \(SJR Q3\).](#)

Tianjin Port/LiDAR-Based Hatch Localization

2021.07 – 2022.04

Data Analyst

Tianjin, China

- Used laser to collect information about shipment and stockpile and simultaneously created a 3D model using Point Cloud.
- Applied Point Cloud and MATLAB to complete the rasterization process, one of the three stages of accomplishing the hatch localization
- After scanning a reference hatch with size 23.55*15.35m using LiDAR, the hatch localization algorithm was able to measure the hatch accurately with the same dimensions, satisfying the anti-collision requirement of error margins less than 0.5 meters.
- Publication: [Zeyi Jiang, Xuqing Liu, Mike Ma, Guanlin Wu, and Jay A. Farrell. "LiDAR-Based Hatch Localization." *Remote Sensing* 14, no. 20 \(2022\): 5069 \(JCR Q1, SCI Q2\).](#)

COURSE PROJECTS

Operating System Course Projects

2022.09 – 2022.12

Independent Projects

CUHK(SZ)

- **Learned about Kernel and loading a module into the Kernel:** Created a kernel thread and forked a process to execute the test program.
- **Used multithread programming to accomplish a game called “Frog crosses river” and implemented a thread pool:** This game was a multithreaded game that required forking threads for its implementation. The thread pool was one that acted as a pool consisting of several threads, which performed the job after a user's request.
- **Implemented the simulation of a mechanism of virtual memory via GPU's memory (CUDA Programming):** Both physical memory and secondary memory were implemented using shared memory and global memory in CUDA. The memory replacement policy for page faults was implemented. The mapping between virtual address (VA) to the physical address (PA) was implemented, and a paging system with swapping between shared memory and global memory was established.
- **Implemented the simulation of a mechanism of file system management via GPU's memory (CUDA Programming):** Implemented the logical file system that managed the metadata information, which included all of the file-system structure except the actual data. Implemented the free-space manager that tracked the unallocated blocks and provided them when needed. Implemented a file-control block (FCB) which contained the information about the file, including ownership, permissions, and location of the file contents.

Design and Analysis of Algorithm Course Project: Pandemic Control Algorithm

2022.09 – 2022.12

Group Leader

CUHK(SZ)

- Implemented both the subtree-heavy algorithm and degree-heavy algorithm and conclude that subtree heavy performs 25% better than the other on average.
- Used Approximation algorithm to prove the ratio between Subtree-heavy Algorithm and Optimal solution, which is

$$S_{greedy} \geq \frac{1}{2} S_{optimal}$$

Database System Course Project: E-commerce Web Shopping Platform

2022.02 – 2022.05

Group Leader

CUHK(SZ)

- Developed a general understanding of the online shop platform.
- Utilized Java, Vue, Maven, and MySQL through the project.
- Implemented the CRUD operations through front-end with corresponding changes to back-end.
- Cultivated leadership and the ability of distributing tasks to members of the group.

Computer Architecture Course Projects

2022.01 – 2022.05

Independent Projects

CUHK(SZ)

- **Implemented a C++ translator for MIPS Instructions set to machine code:** This translator scanned the input file twice and returned a label table and a vector filled with each line of instructions. After processing, it wrote the translated machine code into a .txt file.
- **Implemented a C++ simulator for the memory and register of the computer when running a MIPS asm file:** This simulator displayed the changes in the Stack segment, Data segment (Dynamic data & Static data), and Text segment while running a MIPS code.
- **Implemented an Arithmetic and Logic Unit(ALU) by using Verilog:** This program implemented a simple CPU that supported basic instruction parsing, register value fetching, and ALU functions such as add, addi, etc.
- **Implemented a 5-stage pipelined CPU by using Verilog:** In the past version of this program, there were 5 clock cycles required to execute one instruction. In the first clock cycle, the CPU read one instruction from the instruction memory using the newly given PC address. In the second cycle, the processor divided the instruction into different parts and decoded the MIPS instruction using the registers and control unit. In the third cycle, the ALU handled arithmetic, logical, shifting, and conditional branch instructions. In the fourth cycle, data was fetched from or stored to the data memory by data transfer instructions. In the fifth cycle, data was written back to registers if needed.

AWARD

- School of Data Science **Dean's List** in 2022-2023

Others

- **Language:** English (TOEFL 108; GRE 326+4), Chinese (native)
- **Computer skills:**
 - **Programming Languages:** C, C++, Java, Python, MySQL, MATLAB, Verilog
 - **Frameworks:** PyTorch, CUDA, SpringBoot, Vue
 - **Simulation Platforms and Packages:** ROS, Gazebo, PX4, Rviz, Gmapping, QGroundControl, PointCloud, OpenTCS
 - **Toolchains:** Visual Studio Code, IntelliJ IDEA, PyCharm, CLion, Wenstorm, MATLAB, Conda, Git, LaTeX, Markdown
- **Interests:** Philosophy, RAP, Work out, Basketball